AI Failures Museum: Learning from AI's Failures in Environmental Decision-Making

Group Java_The_Hutt

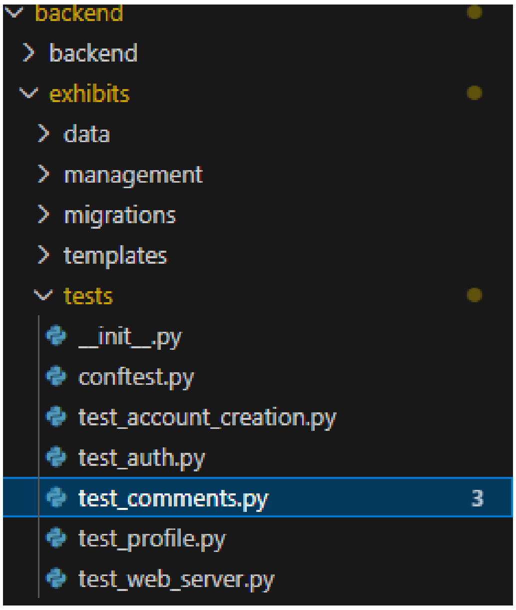
Testing Evidence

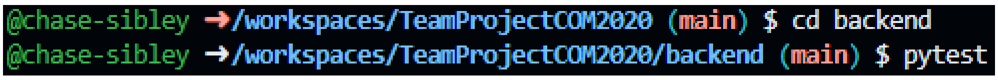
How to run tests:

Tests are stored in the backend of the source code.

```
backend
| → exhibits
    | → tests
```

Tests are ran using the pytest framework. To run, go into the backend directory and run `pytest` in the terminal.





Testing

Areas of the AI exhibit that have been tested are:

- Web server health
  - Endpoints are running
  - Expected response codes
- Account creation and management
  - Creating a valid account with no error
  - Error handling
  - Deleting and modifying user credentials
- Commenting
  - Adding comments to exhibits
  - Replying to comments
- Authentication
  - Expected 302 response codes to actions

Outcomes:

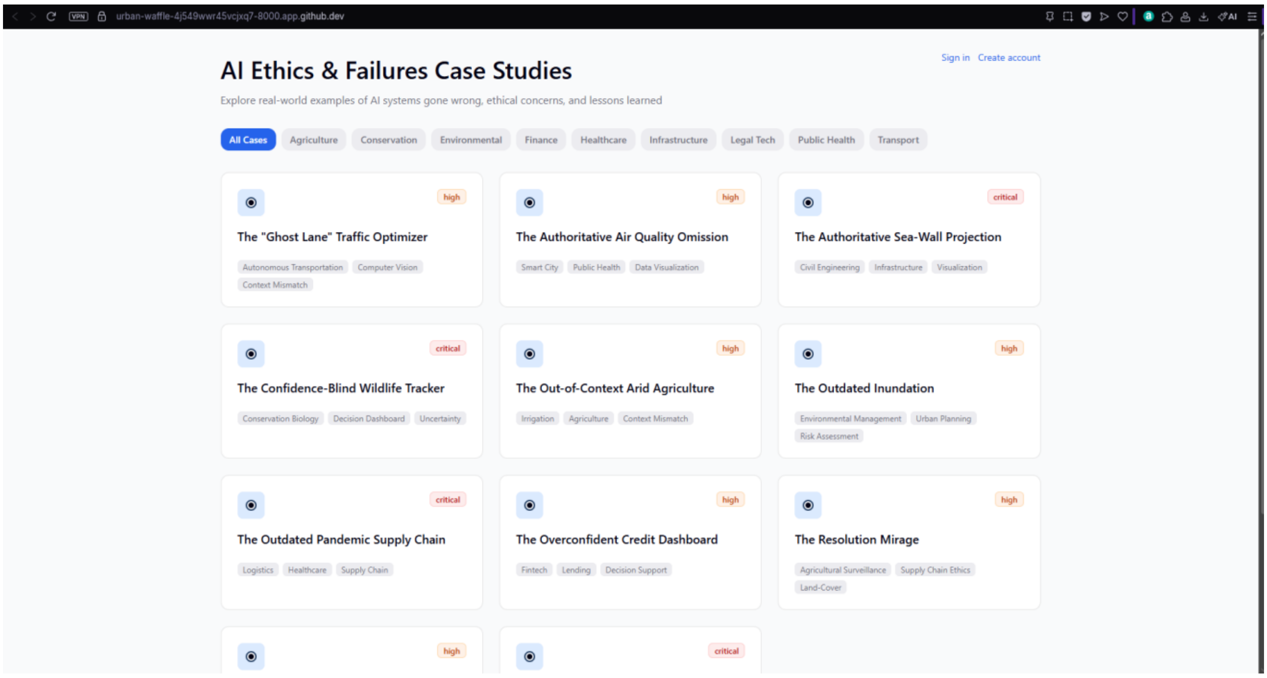
```
rootdir: /workspaces/TeamProjectCOM2020/backend
configfile: pytest.ini
plugins: django-4.11.1
collected 28 items

exhibits/tests/test_account_creation.py ....
exhibits/tests/test_auth.py ...........
exhibits/tests/test_comments.py .....
exhibits/tests/test_profile.py ..
exhibits/tests/test_web_server.py ......
```

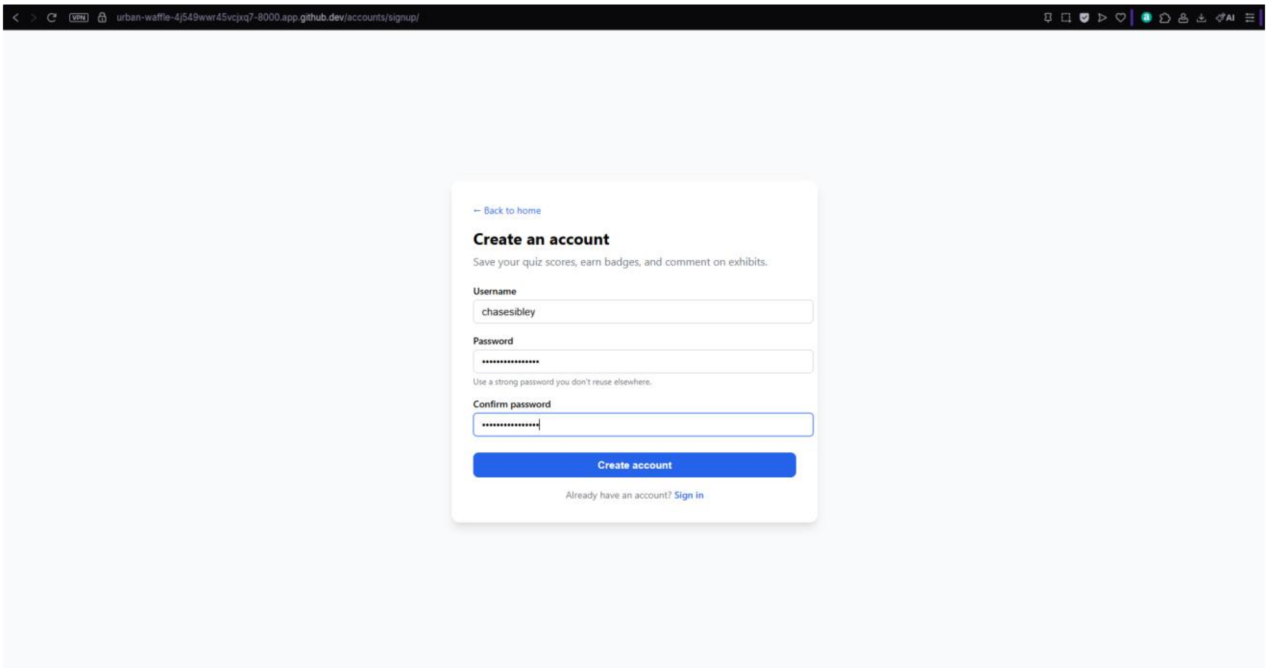
The tests were created to ensure functionality within the exhibit and to make sure that error handling is managed efficiently.

End-to-end example

Home page:



Sign-up:



Signed-in Homepage

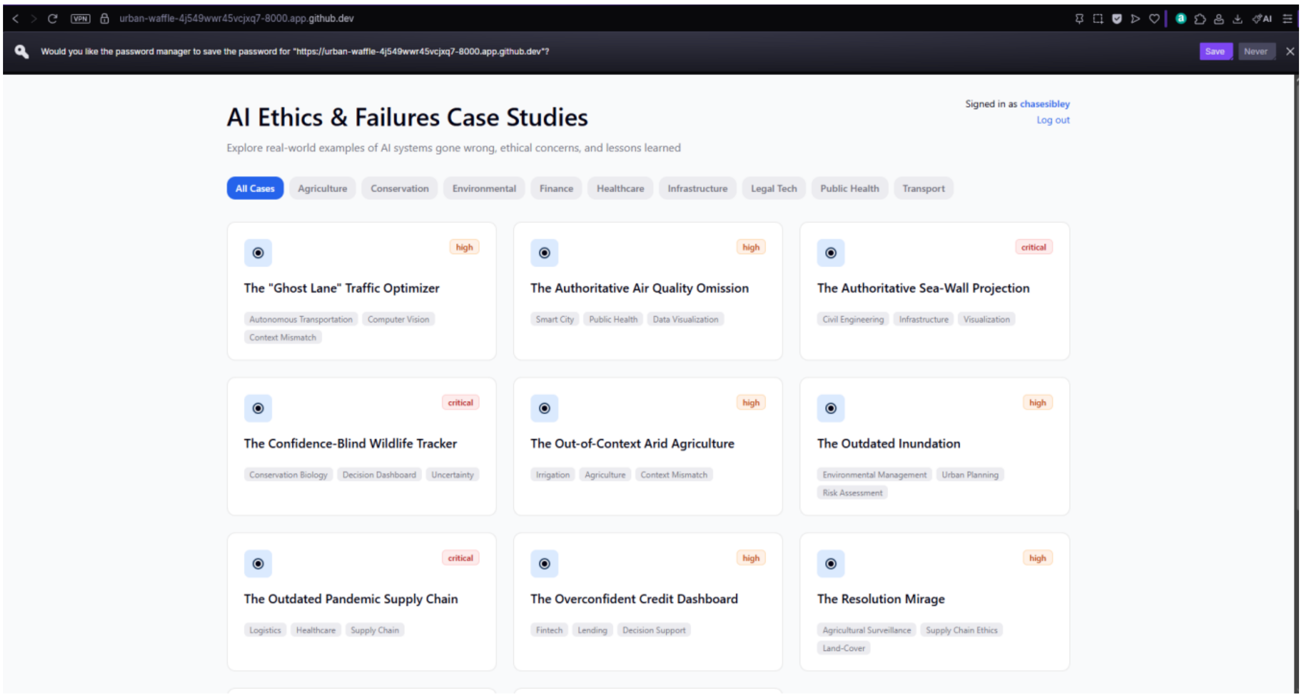
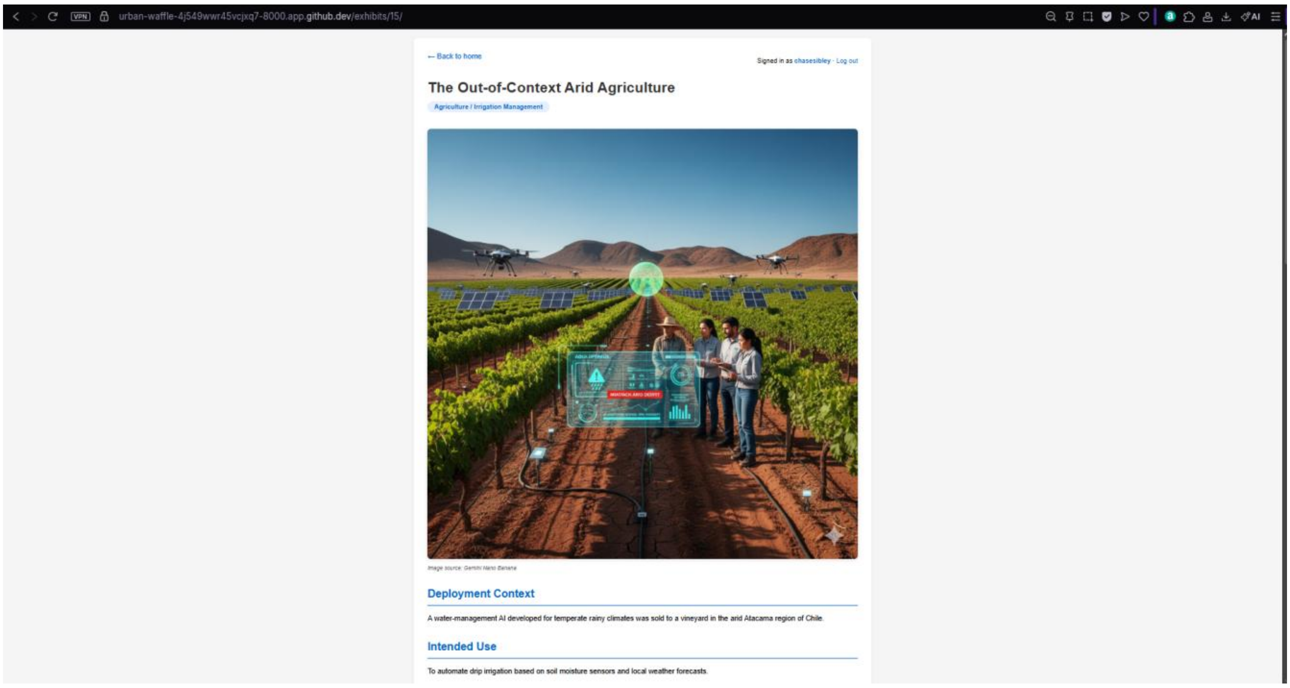
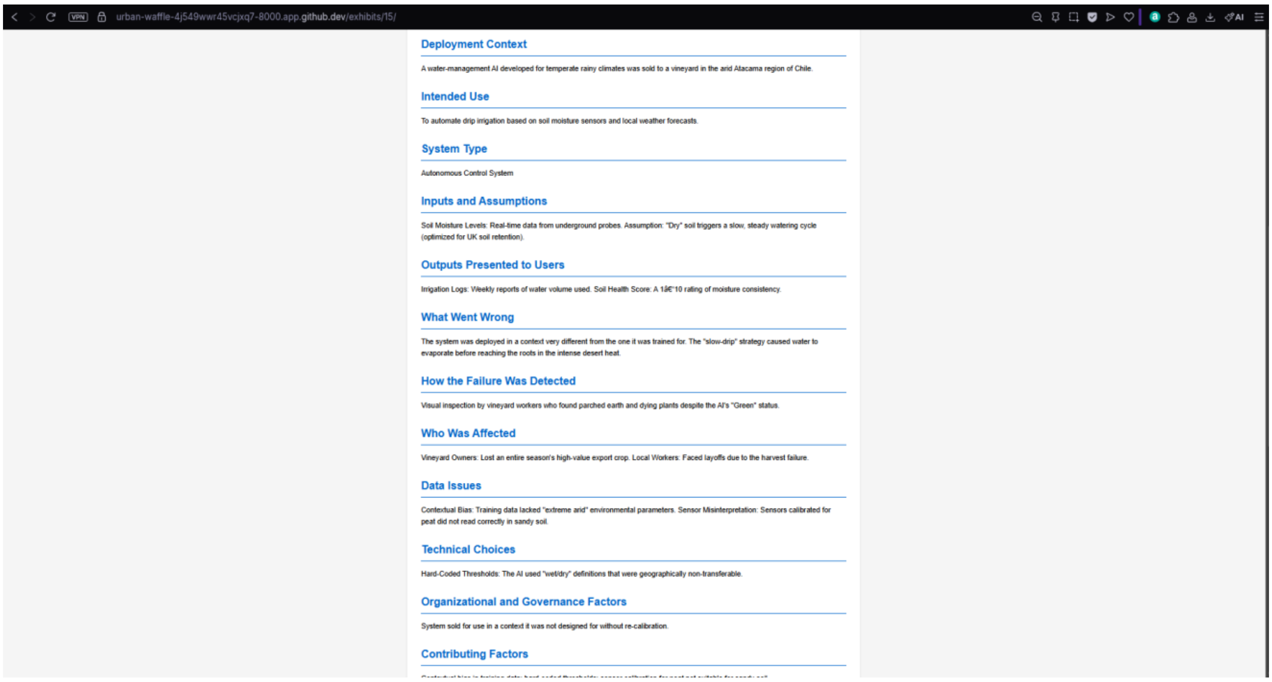


Exhibit View:





urban-waffle-4j549wwr45vcjxq7-8000.app.github.dev/exhibits/15/

Deployment Context

A water-management AI developed for temperate rainy climates was sold to a vineyard in the arid Atacama region of Chile.

Intended Use

To automate drip irrigation based on soil moisture sensors and local weather forecasts.

System Type

Autonomous Control System

Inputs and Assumptions

Soil Moisture Levels: Real-time data from underground probes. Assumption: "Dry" soil triggers a slow, steady watering cycle (optimized for UK soil retention).

Outputs Presented to Users

Irrigation Logs: Weekly reports of water volume used. Soil Health Score: A 1â€"10 rating of moisture consistency.

What Went Wrong

The system was deployed in a context very different from the one it was trained for. The "slow-drip" strategy caused water to evaporate before reaching the roots in the intense desert heat.

How the Failure Was Detected

Visual inspection by vineyard workers who found parched earth and dying plants despite the AI's "Green" status.

Who Was Affected

Vineyard Owners: Lost an entire season's high-value export crop. Local Workers: Faced layoffs due to the harvest failure.

Data Issues

Contextual Bias: Training data lacked "extreme arid" environmental parameters. Sensor Misinterpretation: Sensors calibrated for peat did not read correctly in sandy soil.

Technical Choices

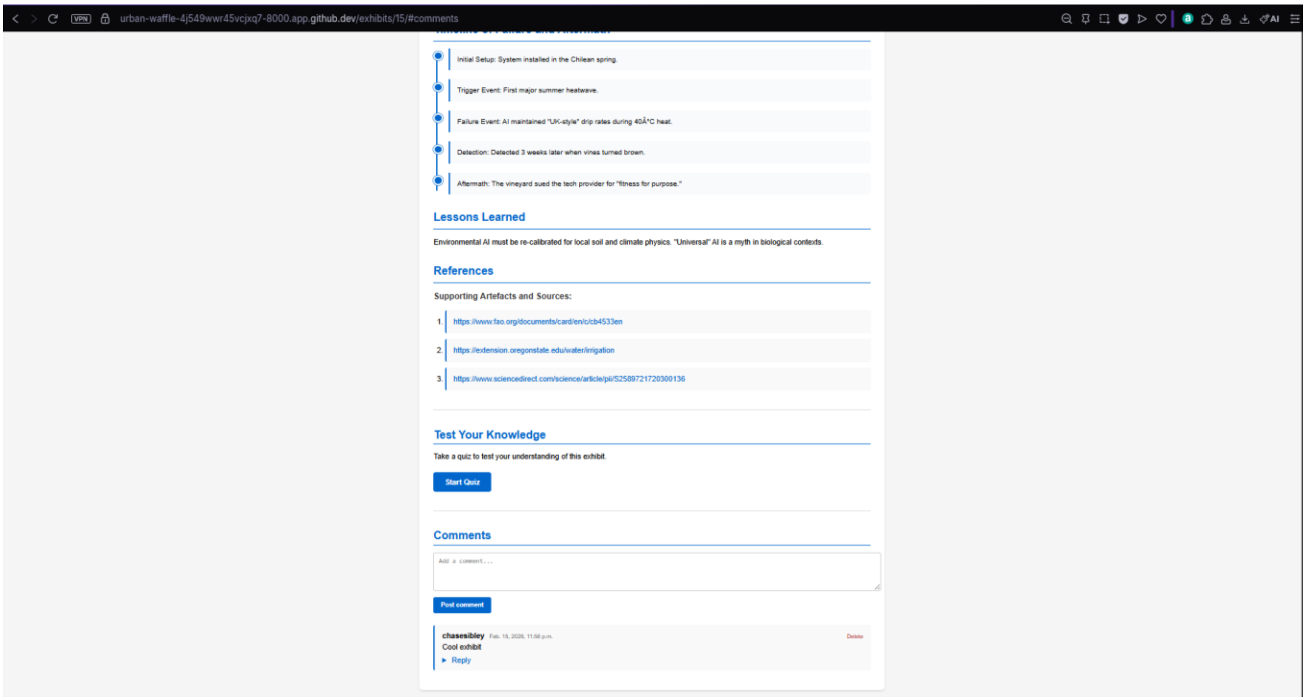
Hard-Coded Thresholds: The AI used "wet/dry" definitions that were geographically non-transferable.

Organizational and Governance Factors

System sold for use in a context it was not designed for without re-calibration.

Contributing Factors

Posting a comment:



urban-waffle-4j549wwr45vcjxq7-8000.app.github.dev/exhibits/15/#comments

- Initial Setup: System installed in the Chilean spring.
- Trigger Event: First major summer heatwave.
- Failure Event: AI maintained "UK-style" drip rates during 40Â°C heat.
- Detection: Detected 3 weeks later when vines turned brown.
- Aftermath: The vineyard sued the tech provider for "fitness for purpose."

Lessons Learned

Environmental AI must be re-calibrated for local soil and climate physics. "Universal" AI is a myth in biological contexts.

References

Supporting Artefacts and Sources:

1. https://www.fao.org/documents/card/en/c/cb4533en
2. https://extension.oregonstate.edu/water/irrigation
3. https://www.sciencedirect.com/science/article/pii/S2589721720300136

Test Your Knowledge

Take a quiz to test your understanding of this exhibit.

Start Quiz

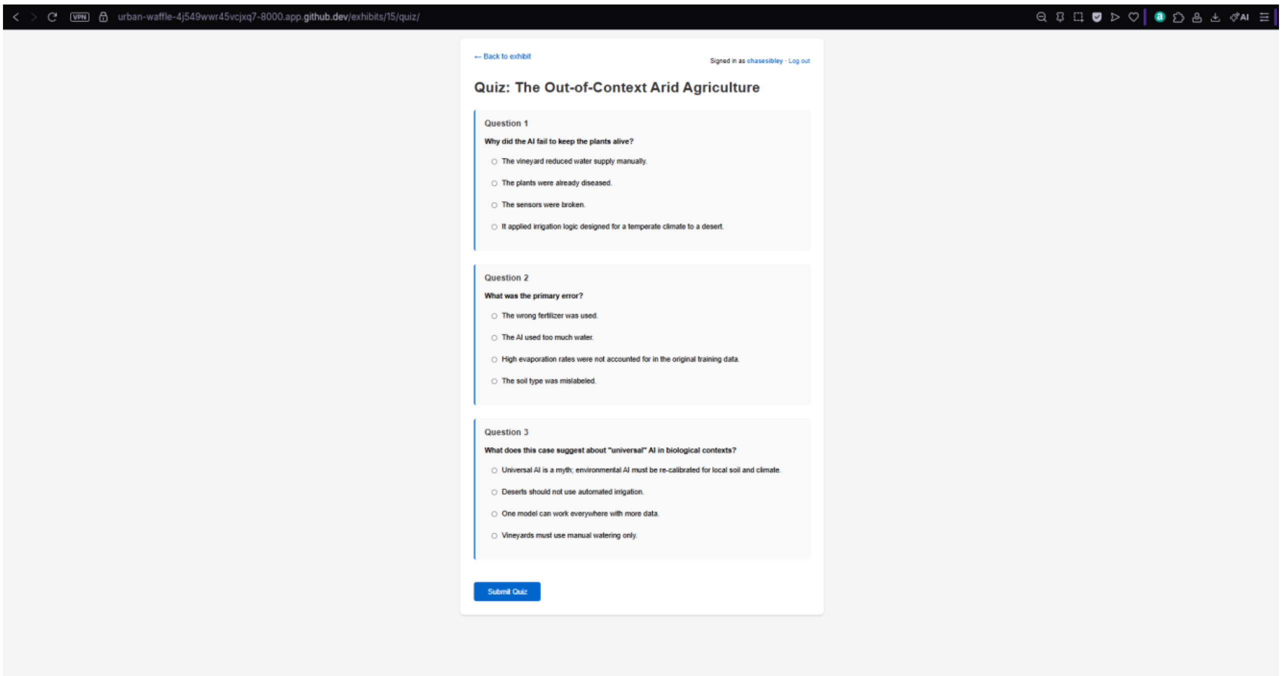
Comments

Add a comment...

Post comment

chasesibley Feb. 15, 2026, 11:56 p.m. Delete
Cool exhibit
► Reply

Quiz



Completed Quiz